Excel XML Converter

User Guide

NMDBiotic & Acoustic conversion pipeline

LPA – 2026 : by KMZ

1. Overview

The application is a Shiny app launched as a single window with 4 tabs (one converter each).

#	Tab	Purpose	Final output
1	BioData_To_Biotic.xls	Build the 4 standard LPA Excel sheets from raw field files	Missions_LPA.xlsx, Fishstations_LPA.xlsx, Catchsamples_LPA.xlsx, Individuals_LPA.xlsx
2	Biotic.xls_To_Biotic.xml	Convert the consolidated Biotic Excel into NMDBiotic v3.1 XML	Biotic_LPA.xml
3	NASC.xls_To_Acoustic.xml	Build the 6-sheet Acoustic_LPA.xlsx from raw NASC export	Acoustic_LPA.xlsx
4	Acoustic.xls_To_Acoustic.xml	Convert Acoustic_LPA.xlsx into Acoustic XML	Acoustic_LPA.xml

2. Tab 1 — BioData_To_Biotic.xls

This tab is itself split into 4 internal sub-modules, each producing one of the LPA Excel files. They can be run independently.

2.1 Fishstations_LPA

Workflow: ① Create Metadata → ② Import Stations file → ③ Inject Metadata → ④ Download Fishstations_LPA.xlsx → ⑤ Download Missions_LPA.xlsx.

Mission metadata (filled manually in the editable table)

- missiontype, startyear, platform, missionnumber
- missiontypename, callsignal, platformname, cruise
- missionstartdate, missionstopdate, purpose

Stations input file — required columns

Column names are normalized to lowercase and underscores; accents are stripped automatically.

Column	Meaning
date	Station date (any common format)
station	Station number
heure_filage	Shooting time (HH:MM or HH:MM:SS)
heure_virage	Hauling time
latitude_filage, longitude_filage	Start position (decimal)
latitude_virage, longitude_virage	End position (decimal)

Column	Meaning
prof_fond_filage, prof_fond_virage	Bottom depth start/stop
prof_filet_filage, prof_filet_virage	Net (gear) depth start/stop
duree_chalutage	Soak time
longueur_funes	Wire length
vitesse_chalutage	Vessel speed
cap_chalutage	Direction (heading)
note	Free comment

Note: missing position columns are flagged with a warning but do not block the export. Empty positions are highlighted in yellow in the preview.

2.2 Catchsamples_LPA

Import a single catches file (CSV / XLSX). Column names are detected with fuzzy matching (slugs + aliases). The following logical columns are required:

Logical name	Accepted aliases (examples)
Species	espece, espèce, esp, scientificname
Station	station, serial, serialnumber
Sample weight	Ech_P, ech_p, poids_ech
Total count	Tot_N, totn, n, n_tot
Total weight (optional)	Tot_P, totp

A WoRMS validation modal appears for any unrecognized scientific name (referential file expected in data/).

2.3 Individuals_LPA

Import two files: a length frequency file and a biology file.

Length frequency file — required columns

```
station, nombre, poids_mens, espece, taille_cm
```

Biology file — required columns

```
espece, station, nombre, poids_g, sexe, sms
```

Join key: same station + same espece; rows are matched by ascending order of nombre. The sexe field accepts M / F / I (mapped to 1 / 2 / NA).

2.4 Biotic_LPA

Import the four files produced in 2.1–2.3 (Missions_LPA, Fishstations_LPA, Catchsamples_LPA, Individuals_LPA). The module assembles them into a single workbook Biotic_LPA.xlsx containing the four sheets named exactly: missions, fishstations, catchesamples, individuals.

3. Tab 2 — Biotic.xls_To_Biotic.xml

Input: the file Biotic_LPA.xlsx produced by Tab 1.

Mandatory sheets (exact names)

Sheet	Required key columns
missions	missiontype, startyear, platform, missionnumber, missiontypename, callsignal, platformname, cruise, missionstartdate, missionstopdate, purpose
fishstations	serialnumber (used as XML attribute) + all standard NMDBiotic station fields
catchesamples	serialnumber, catchsampleid
individuals	serialnumber, catchsampleid, specimenid

The XML is generated under namespace http://www.imr.no/formats/nmdbiotic/v3.1 and downloaded as Biotic_LPA.xml.

All cells are read as text (col_types = "text") — date/numeric formatting must already be correct in the workbook.

4. Tab 3 — NASC.xls_To_Acoustic.xml

Input: raw NASC export, .csv or .xlsx.

Required columns (validated; the app blocks with a modal if any is missing)

```
Date_S, Time_S, Date_E, Time_E, PRC_NASC, Lat_S, Lon_S, Lat_E, Lon_E,
Exclude_below_line_range_max, Exclude_below_line_depth_min,
Exclude_above_line_depth_mean, Exclude_below_line_range_mean, Species_percent,
Species_name, Layer, Interval
```

Manual metadata (sidebar inputs)

report_time, lsss_version, nation, platform, cruise.

QC rules applied automatically

- Rows with Lat_* or Lon_* equal to 999 or -999 are flagged.
- Rows with Sa = 0 are flagged.

Species recognized for acocat mapping (any other value → 16 = Autres)

acocat	Species_name
1	Sardina pilchardus
2	Engraulis encrasicolus
3	Scomber colias
4	Trachurus trachurus
5	Trachurus trecae
6	Trachurus picturatus

acocat	Species_name
7	Trachurus mediterraneus
8	Sardinella aurita
9	Sardinella maderensis
10	Macroramphosus scolopax
11	Macroramphosus gracilis
12	Scomber scombrus
13	Caranx rhonchus
14	Plancton
15	Mesopelagique
16	Autres

Output

Acoustic_LPA.xlsx with 6 sheets — root_metadata, distances, frequencies, ch_types, sa_by_acocat, sa. The 6-tab preview is displayed automatically after processing.

5. Tab 4 — Acoustic.xls_To_Acoustic.xml

Input: the Acoustic_LPA.xlsx produced by Tab 3 (the 6 sheets above).

The app reads every sheet, generates the corresponding Acoustic XML structure, and provides a live XML preview pane plus a download button delivering the final Acoustic_LPA.xml.

No additional column is required beyond what Tab 3 already produced.

6. End-to-end pipeline

```
Raw stations / catches / length / biology files        ↓ (Tab 1) Missions_LPA +
Fishstations_LPA + Catchsamples_LPA + Individuals_LPA        ↓ (Tab 1 – Biotic
module) Biotic_LPA.xlsx        ↓ (Tab 2) Biotic_LPA.xml   ☑ NMDBiotic v3.1  Raw
NASC export        ↓ (Tab 3) Acoustic_LPA.xlsx (6 sheets)        ↓ (Tab 4)
Acoustic_LPA.xml  ☑
```

7. Common pitfalls

- Sheet names are case-sensitive in Tab 2 (missions, fishstations, catchesamples, individuals).
- Tab 1 normalizes column names (lowercase, no accents, underscores). "Latitude Filage" and "latitude_filage" are equivalent; "Lat. filage" is not.
- Tab 3 expects the exact column names listed above (case-sensitive).
- Coordinates encoded as 999 or -999 are treated as missing during QC.
- The WoRMS reference file must be present in data/ — otherwise species validation is skipped with a warning.
- All four tabs share the same Shiny session: do not open the same inputId twice across tabs.

LPA – 2026 : by KMZ